NUI Galway
OÉ Gaillimh

Semester I Examinations 2019/2020

Exam Code(s)	1SD3, 1SD1, 1MF1
Exam(s)	Higher Diploma in Software Design and Development MSc in Software Design and Development
Module Code(s)	CT537
Module(s)	Software Engineering Methods
Paper No.	1
Repeat Paper	
External Examiner(s)	Prof. Pier Luca Lanzi
Internal Examiner(s)	Prof. Michael Madden Ms. Karen Young*

Instructions: Candidates should answer **any 3** questions.

Duration	2 hours
No. of Pages	5
Discipline(s)	Computer Science
Course Co-ordinator(s)	Dr. Enda Barrett, Dr. Seamus Hill

Requirements:

Release in Exam Venue	Yes	X	No	
MCQ	Yes		No	X
Handout	None			
Statistical/ Log Tables	None			
Cambridge Tables	None			
Graph Paper	None			
Log Graph Paper	None			
Other Materials	None			
Graphic material in colour	Yes		No	x

1.(a) Construct a **Class Diagram**, illustrating the relevant **domain classes** (attributes and operations), and their **relationships** (associations, inheritance) based on the textual description of a *University Library* system presented below:

University staff and students are usual users of the library. The library system records the books and journals owned by the library and records loans to members. Each member has a unique ID number. In order to borrow a book, the library member must show a valid ID card, which is checked to ensure that it is still valid against the student, staff or guest databases maintained by the university. The system must also check to ensure that the borrower does not have any overdue books or unpaid fines before he or she can borrow another book.

The library maintains a large collection of borrowable items (e.g. books, cds, etc.) which it continuously updates, both automatically when new editions become available and as a result of staff requests. There are reserved items that can only be borrowed by staff for one day. There are also seven days and 28 days loan items that can be borrowed by both staff and students. Sometimes books are lost or are returned in damaged condition. The manager must then remove them from their stock records and will sometimes impose a fine on the borrower. Library members can also request to borrow items from other libraries via an inter-library loan system managed by the library.
Staff members can borrow up to 8 items at a time, and students can borrow a maximum of 3 items. Librarians run the library and provide services (e.g. managing membership, lending items) to the members. Each librarian has a unique login name and has to login to the system before using it. Every week, the library contacts (by e-mail or text message) those people with overdue books. If a book is overdue by more than two weeks, a fine will be imposed and a librarian will telephone the borrower to remind him or her to return the book(s).

(15)

(b) You have been asked to select an appropriate **process model** for the development of the above library project. Evaluate your alternatives, identifying their strengths and weaknesses, and select one, clearly outlining your evaluation criteria and the reasons for your selection.

(5)

2. (a) Prepare a **Use Case Diagram** for the *Arts Festival* system described below and expand ONE of the use cases to a full textual description.

> You have been asked to design an interactive system (to include a mobile application) to support the city's Arts Festival. The system should provide information on all Arts Festival events: date, location, producer, performers, reviews etc. In addition, the system must support ticket bookings and payments, seating selection and provide interactive demos and snippets from the performances along with user reviews. Opportunities to get involved in the interactive digital performances or play "games" associated with the festival should also be supported by this system. Management can run reports on bookings and sales as well as review system activity relating to interactive demos, digital performances and associated games.

(10)

(b) Prepare an **Interaction (sequence)** diagram for the expanded Use Case for the *Arts Festival* system above.

(10)

3. (a) Prepare an **Activity Diagram** for the *Airline Reservations System* described below, using swimlanes if required to illustrate the different processes:

> OO Airlines runs flights from Java Valley. The reservation system keeps track of passengers who will be flying in specific seats on various flights, as well as people who will form the crew. OO Airlines runs several daily numbered flights on a regular schedule (requires date, time and flight number: both planned and actual as flights can run late). These flights fly from one airport terminal to another, including any intermediate stops. OO Airlines operate a frequent-flier plan for passengers: passengers are automatically part of a frequent-flier plan. They accumulate points for every flight they take with OO Airlines.
> This system needs to support the creation and cancellation of flights with associated crew, searching for particular flights, passenger cancellations, passenger queries to determine what flights and flight times are available, and the prices of flights (the airline publishes prices that apply between any pair of cities to which it flies).

(8)

PTO

(b) Experienced software developers often say, "*Testing never ends, it just gets transferred from you [the software engineer] to your customer.*"

Assume that you are developing a new online social networking app. The app will provide typical social communication functions, but also maintains a record on each customer. Outline a ***test strategy*** for this system that will ensure comprehensive testing of the system before "customer tests" begin.

(8)

(c) Evaluate the contribution of software engineering techniques to improving the **quality** of software delivered.

(4)

4. (a) Construct a **State Transition Diagram** to model the below description of a Car Wash Control Programme:

A car wash has a software system to monitor and control its activities.
A control pedestal accepts money and provides instructions for the kind of wash to do. When no car is in the washer, the pedestal is placed in a resting state where it displays a message inviting customers to deposit money. The pedestal accepts coins, notes and credit cards. As money is deposited, the pedestal displays the total deposited so far. If the customer presses a coin return lever, all deposited money is returned and the pedestal returns to its resting state.
If enough money is deposited, the pedestal will respond to button presses for a regular wash (5 Euros), or a super wash (7 Euros). The pedestal will then deliver change, if necessary, and display a message telling the customer to enter the washer.
When a car enters the washer, a sensor sends an event that alerts the system to have the pedestal display a message for other customers to wait and then run the appropriate wash cycle. During this time, the pedestal will not accept any money or attend to any button presses.
There is also a sensor to detect when a car leaves. If a car leaves before the cycle is done, the cycle is stopped and the pedestal is returned to its resting state. Otherwise, when the cycle completes, the system waits for the car to exit the washer. When the care exits the washer, the pedestal is returned to its resting state.

(10)

(b) You are the team lead of a systems design team that is implementing a multiple module project with an expected duration of one year. Your manager just sent you the memo included below. Write a memo back to your manager answering their questions.

> *Dear Sam,*
>
> *I was speaking with Tom Dufour last week and he mentioned that many of our teams are now using the code generation facilities in our CASE technology to speed up delivery times. Will this save us time and money do you think? Are you aware of any problems we might encounter if we use this approach?*
>
> *Regards,*
> *XYZ.*

(6)

(c) Describe the concept of software process maturity in your own words. What different **SPI (Software Process Improvement)** frameworks are you aware of? What are the key organisational characteristics needed to successfully improve an organisation's software process?

(4)